

Aneshka Moudry

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Education

Department of Physics, University College London & Phasecraft

Sep 25 — Sep 29

PhD, Quantum Computation & Communications

- Designing novel algorithms to simulate condensed matter physics and chemistry on quantum computers.
- Fully sponsored and supervised by [Phasecraft](#) [🔗](#), a leading quantum algorithms company.
- *Modules*: Quantum many-body physics, tensor networks, quantum algorithms, computational complexity, quantum information theory, error mitigation and error correction, hardware platforms for quantum computation.

Department of Chemistry, St Peter's College, University of Oxford

Sep 21 — Jul 25

Integrated master's (MChem), Chemistry

- Ranked 2/197 in cohort, with overall first-class honours (average of 83%).
- *Examinations*: Thesis - 80%, Final examinations - 85%, Second year - 79%, First year - 79%, Laboratory work - 77% (equivalent to 4.0 GPA).
- *Key modules*: Electronic structure theory, advanced quantum mechanics, solid state systems, computational statistical mechanics, group theory, calculus and linear algebra, thermodynamics.

Research Experience

Quantum Algorithms Researcher, Phasecraft & UCL

Present

- *Thesis*: Simulating quantum materials on quantum hardware - designing novel quantum algorithms to approximate time dynamics and dynamical correlation functions.
- Expertise in numerical condensed matter physics, Julia and Python coding, computational complexity, linear algebra, and quantum mechanics. Experience with running experiments on real quantum hardware and classical emulations.
- UCL projects:
 - Methods for the decomposition of classically computed matrix-product states into quantum circuits (Prof. Andrew Green, Physics).
 - Neural network quantum states for classical simulation of quantum circuits (Prof. Mehrnoosh Sadrzadeh, CS).

Master's thesis, Physical and Theoretical Chemistry Lab, Oxford

Oct 24 — Jul 25

- *Thesis*: 'Theoretical excited-state dynamics of substituted molecular photoswitches' (publication in progress), computational quantum chemistry, [Kirrander group](#) [🔗](#).
- Developed an object-oriented data extraction framework in Python to analyse over 2TB of data on an HPC cluster, coupled to a high-level clustering algorithm software stack for data analysis.
- *Skills*: Deep knowledge of theoretical quantum chemistry and experience with modern computational methods to numerically approximate electronic wavefunctions. Extensive coding in Python and Fortran, operating HPC clusters, Unix environments, and shell scripting to automate computational pipelines. Clustering and ML for large dataset analysis.

Work experience

Private Tutor

Aug 23 — Current

- Tutored and home-schooled A-level (and equivalent) and GCSE mathematics, physics, chemistry and biology to over 50 students (roles at OXED, Tutor Doctor, Casale Tutoring, and Inspire Education), both in-person and online.
- Extensive experience using online whiteboard-based platforms, eg. Zoom, Bramble, Teams, Miro, to augment digital learning.
- Supplemented personalised 1 hour tutorial sessions with individually tailored revision plans, homework, mock examinations and summary notes to foster external learning and maximise content retention.

Global Markets Trading Summer Intern, Bank of America

Jun 24 — Sep 24

- Independently researched and developed quantitative investment strategies for algorithmic options trading desks, and presented this to senior members of the bank.
- Application of interpretable machine learning algorithms in Python for statistical analysis of large, incomplete time series datasets, developing extensive problem-solving and analytical skills.

Venture Capital Intern, Future Planet Capital

Aug 23 — Mar 24

- Provided scientific consultation and due diligence of deep tech companies to assist partners with investment decisions, following thorough analysis of markets, scientific papers, and financial statements.
- Construction of investment memoranda recommending investment in two deep tech startups. Awarded best presentation pitch and selected as top intern from a cohort of 25.

Mathematics, Physical and Life Sciences Divisional Representative, Oxford SU

2023

- Furthered excellent communication and collaboration skills when representing 2600+ UG students at university board meetings and through Oxford SU council voting. Extensive experience with academic policy papers, and university-wide data collection and analysis.

Data scientist, Oxford Digital Strategy Group

Apr — Jul 22

- Development programme in Python-based data analysis (pandas), extracting insight from big data sets for digital consultancy projects. Enhanced quantitative business analysis and presentation skills through three team-based case studies.

Skills

Programming languages: Python (NumPy, pandas, SciPy, scikit-learn), Julia, Fortran, MATLAB, Bash, L^AT_EX.

Software: Git/GitHub, Linux environments, operating HPC systems, MS Office (Word, PowerPoint, Excel inc. VBA).

Languages: English (Native), Czech (Fluent), German (Intermediate), Spanish (Elementary).

Citizenship: British, American.

Prizes

Thesis Prize in Theoretical and Physical Chemistry

Awarded for exceptional performance in the fourth year research thesis.

University of Oxford

October 2025

Gibbs Prize in Chemistry

Awarded for exceptional performance in final year examinations.

University of Oxford

August 2024

Shimadzu Prize in Chemistry

Best performance in practical chemistry and laboratory work in the bachelor's degree program.

University of Oxford

August 2024

Domus Scholarship

Awarded for first-class results in preliminary examinations.

University of Oxford

August 2022

Bailey Scholarship

Awarded for best performance in the college cohort in preliminary examinations.

University of Oxford

August 2022

References

Dr. Raul Santos

Lead Quantum Scientist, Phasecraft, London, United Kingdom

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Scholar Profiles: [Google Scholar](#) 

Prof. Adam Kirrander

Professor of Physical and Theoretical Chemistry, University of Oxford, Oxford, United Kingdom

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